

Aakash GURUNG

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EDUCATION

Expected Dec 2026	M.A. in Mathematics, The University of Alabama, Tuscaloosa, AL
Expected May 2026	B.S. in Mathematics; Minor: Digital, Public, and Professional Writing, The University of Alabama, Tuscaloosa, AL

RESEARCH INTERESTS

I am broadly interested in **Algebraic Combinatorics**, **Applied Mathematics**, and **Geometry**.

RESEARCH EXPERIENCE

Present Mar 2026	LLM-Driven Epidemic Modeling, UNIVERSITY OF ALABAMA, <i>Mentors: Professors Chuntian Wang and Jia Zhao</i> › Distilling LLM Agent Behavior Epidemic ModelingLarge Language Models
Present Feb 2026	Rectifiability, UNIVERSITY OF ALABAMA, <i>Mentor: Professor Simon Bortz</i> › Investigating rectifiability in parabolic setting. Geometric Measure TheoryRectifiability
Present Feb 2025	Combinatorial Perspective of (q, t)-Catalan Symmetry, UNIVERSITY OF ALABAMA, <i>Mentor: Professor Kyungyong Lee</i> › Working on a subset of the combinatorial proof of the (q, t)-Catalan symmetry. Algebraic CombinatoricsCatalan Numbers
Dec 2025 May 2025	Finite-Size Effects in Epidemic Models, UNIVERSITY OF ALABAMA MATHEMATICS SUMMER REU, <i>Advisors: Professors Chuntian Wang, Yuanyuan Song, Yuanzhen Shao</i> › Co-developed agent-based and mean-field SIHRS models incorporating immunity waning to capture recurrent epidemic waves. › Employed a martingale-based early-time-step method to identify non-linear amplification of finite-size effects. › Validated theoretical predictions via numerical simulations calibrated to county-level COVID-19 data. Applied MathematicsEpidemic ModelingStochastic Analysis
June 2024 Feb 2024	Game of Cycles on Maximal Plane Graphs, CUNY RESEARCH SCHOLARS PROGRAM, <i>Advisor: Professor Malgorzata Marciniak</i> › Defined “IO Maximal Plane Graphs” and analyzed invariant properties to determine the game outcome. › Established that the winning strategy is determined by the parity of the graph’s vertices. Combinatorial Game TheoryGraph Theory
Aug 2023 June 2023	Method of Brackets and Bessel Function Integrals, POLYMATH JR 2023, <i>Advisor: Professor Victor H. Moll</i> › Applied the “Method of Brackets” to provide rigorous proofs for entries involving Bessel functions of the first and second kind from the Gradshteyn and Ryzhik tables. Special FunctionsIntegral CalculusBessel Functions
Aug 2023 Mar 2023	Continued Fractions, a-Fibonacci Numbers, and Middle b-Noise, INDEPENDENT PROJECT, <i>Advisor: Professor Cheng Han Pan</i> › Generalized palindromic continued fractions $[1, \dots, 1, 3, 1, \dots, 1]$ to $[a, \dots, a, b, a, \dots, a]$ using a-Fibonacci sequences. › Showed that the a-th metallic ratio limit is invariant under the middle noise term b. Number TheoryContinued FractionsFibonacci Sequences

PREPRINTS

Submitted	A. Gurung, S. Wagle, A. Carr, C. McCann, K. Kodatt, Y. Song, Y. Shao, C. Wang. “An exploration of finite-size effects in the dynamics of epidemic compartmental modelling.”
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EXPOSITORY/OTHER WORKS

- 2024 **A. Gurung** and C.-H. Pan. "Continued Fractions, a -Fibonacci numbers, and the middle b -noise," *Mathematics Exchange*, 18(1), 77–87.
- 2024 (with the Polymath Jr. Group). "The integrals in Gradshteyn and Ryzhik. Part 34: Bessel functions," *Scientia Series A: Mathematical Sciences*, 34, 109–129.

CONFERENCES & WORKSHOPS

- March 2026 AMS Southeastern Sectional Meeting, **Presenter**
- May 2024 CUNY Undergraduate Research Day 2024, **Presenter**
- Aug 2024 MathFest 2024, **Presenter**
- Sept–Nov 2024 Preliminary Arizona Winter School 2024: Symmetries of Root Systems, Attendee

HONORS & AWARDS

- 2026 Thomas W. Palmer Math Award, University of Alabama
- 2026 Randall Outstanding Undergraduate Research Award, University of Alabama
- 2025 ASSURE Grant, University of Alabama
- 2024 Best Poster Award, CUNY Undergraduate Research Day
- 2023 Samuel J. Steinberger, Jr. Memorial Award, Juniata College
- 2020–2021 Nepal Mathematical Olympiad (Top 12); Nepal Astronomy Olympiad (Rank 1); USA Astronomy & Astrophysics Competition (National Qualifier)

WORK EXPERIENCE

- Present**
Aug 2025 | **IT Service Desk Student Assistant, UNIVERSITY OF ALABAMA, Tuscaloosa, AL**
 > Provide timely software and technology support to resolve user issues efficiently.
 IT Support Technical Support
- May 2025**
Sept 2024 | **Peer Tutor, MATHEMATICS TECHNOLOGY LEARNING CENTER, Tuscaloosa, AL**
 > Drop-in tutor for Calculus 1, 2, and 3.
 > Run recitation classes for Calculus 1.
 Calculus Mathematics Education Tutoring

SKILLS

- Programming** Python (NumPy, Pandas, SciPy), Julia, MATLAB, JavaScript, HTML, CSS
- Tools** PowerQuery, $\text{\LaTeX}$, Git
- Other** Grant Writing